

Virtual Lab by Quantum Flytrap: Interactive simulation of quantum mechanics

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ABSTRACT

Virtual Lab by Quantum Flytrap explores novel ways to represent quantum phenomena in an interactive and intuitive way. It is an online laboratory with a real-time simulation of an optical table, supporting up to 3 entangled particles. We have created a highly visual no-code interface and an introductory Quantum Game to make the Virtual Lab approachable for users without prior exposure to quantum mechanics. At the same time, a configurable sandbox mode with customizable elements makes it useful for students, lecturers, and scientists. Virtual Lab provides a way to explore exact quantum states, entanglement, and the probabilistic nature of quantum physics. We provide ways to simulate quantum computing (e.g. Deutsch-Jozsa algorithm), use quantum cryptography (e.g. BB84 protocol), explore counterintuitive quantum phenomena (e.g. quantum teleportation), and recreate historical experiments (e.g. Michelson-Morley interferometer). Virtual Lab is available at: <https://lab.quantumflytrap.com/>.

CCS CONCEPTS

• Visualization techniques; • Human-centered computing; • Graphical user interfaces;

KEYWORDS

Quantum mechanics, Quantum computing, Simulation

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1 INTRODUCTION

Quantum mechanics is commonly considered a difficult subject. Recently, motivations to understand it went beyond physics education and pure curiosity. Quantum technologies rise to importance at a rapid pace [1], with estimations of more than \$1 trillion value potential by the mid-2030s [2]. Quantum cryptography is already in proof-of-concept use, while quantum computing will likely add business value in the next few years [3]. Consequently, quantum

literacy [4] is becoming more important in order to keep up with technological possibilities and career opportunities.

At Quantum Flytrap, we develop user-friendly interfaces to help beginners, STEM students, educators, and software & quantum professionals learn and use quantum technologies. We aim to bridge the gap between quantum hardware and end-users.

Our tech demo Virtual Lab (available at <https://lab.quantumflytrap.com/>) is a free-to-use online laboratory with a real-time simulation of quantum states of the elementary particles of light: photons. Photons serve as the main resource for quantum communication and as one of the major implementations of quantum computing. For beginners, we aim to lower barriers to entry and serve as their first step to quantum literacy. For educators, the Virtual Lab is an interactive and intuitive teaching aid. For advanced users, we offer a fast and efficient tool for prototyping experiments.

2 CONTEXT

Quantum mechanics is counterintuitive - not only to a general audience but even to its own creators. Neils Bohr claimed that “those who are not shocked when they first come across quantum theory cannot possibly have understood it.” Albert Einstein, perplexed by the curious quantum properties of two and more particles, called it “a spooky action at a distance” - a property now called entanglement. Quantum mechanics and its “many-world” interpretation spark the imagination in popular culture [5] and spirituality [6], but it is rarely presented in a factual, digestible way. A handful of accessible books [7, 8] serve as a good start but still require a fair dose of vector algebra.

We lack everyday exposure to quantum phenomena. Most of them happen on a tiny scale (e.g., electron bonds holding molecules together) and cannot be seen with the naked eye. Fortunately, we can simulate small quantum systems with computers and give access to the fascinating world of quantum. When we are children, we learn the basics of mathematics or classical physics intuitively - by counting fingers and playing with objects. Yet when it comes to quantum mechanics, currently, the very first step is to get acquainted with complex numbers and wave functions.

There are already a few interactive simulations of quantum mechanics. Early attempts from the 2000s include a simulation of wavefunctions [9], i.e., descriptions of a particle in many places at once. Newer ones include simulations of quantum computing circuits - e.g., Quirk [10] and IBM [11] Experience. Most of them, though, requires prior knowledge of quantum physics. A different approach is to use games based on quantum physics [12]. John Preskill writes that “perhaps kids who grow up playing quantum games will acquire a visceral understanding of quantum phenomena that our generation lacks.” [13]

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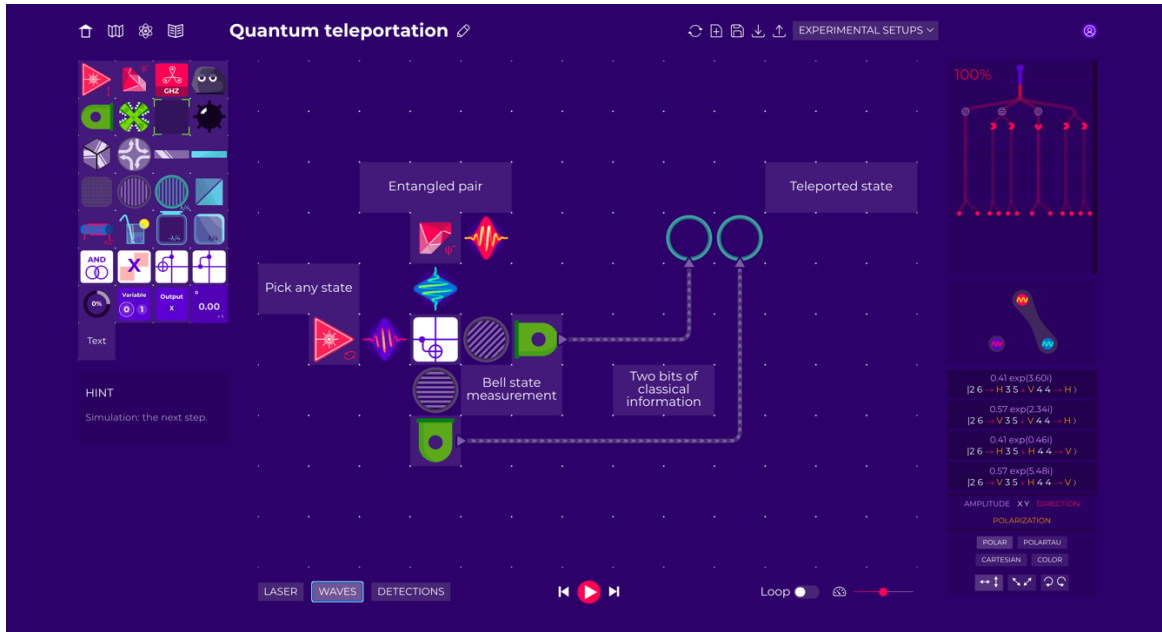


Figure 1: The Virtual Lab, Quantum Teleportation simulation

In particular, this project is the successor of an online open-source puzzle game about quantum physics Quantum Game with Photons (2016) [14], later developed as Quantum Game with Photons 2 (2019) [15] at the Centre For Quantum Technologies at the University of Singapore.

3 DESIGN

We need quantum “LEGO bricks” to allow us all to play and experiment with quantum mechanics by composing arbitrary setups: educational, scientific, puzzles, and interactive arts. We aim to lower the entry barriers by using a highly visual no-code Graphical User Interface (GUI) and delegating all computing to computers. Since our goal is to reach a diverse audience, the Virtual Lab is browser-based. It can be accessed from any place in the world, from a desktop, tablet, or mobile, from any operating system with a modern browser. Additionally, the install-free nature allows users to start using it in seconds. The main content is available without login.

The Virtual Lab can be used to create arbitrary experiments (in the sandbox mode), view and modify historical physics experiments, and play as a game where you solve a series of puzzles. Users can save their experiments and share them publicly via a link.

The project evolved from an online game and we found that the intuitive, game-like visual language was well received, especially by students and lecturers.

The sandbox (Figure 1) is the primary part of the Virtual Lab. Elements, such as photon sources, detectors, mirrors, beamsplitters, or logic gates, are moved to the canvas area with drag & drop mechanisms. Users can customize the elements by configuring their parameters, e.g., the reflectance of mirrors or polarization rotation angle for Faraday rotators. Users can work in two modes:

either seeing the averaged beam of light or tracking single photons step-by-step.

We investigated new and intuitive ways to represent quantum phenomena, including intrinsically quantum phenomena such as superposition and entanglement. Users can investigate quantum entanglement through animated icons of the photons on the canvas (Figure 2 A). In particular, quantum entanglement is represented by the correlated blinking of the particles. Additionally, a blob-like diagram represents the entanglement measure between particles (Figure 2 B). For the most advanced users, we have a viewer allowing exploration of the exact quantum state [16], with a choice of basis and coordinates of complex numbers (Figure 2 C).

There are several interpretations of quantum mechanics, including the indeterministic Copenhagen interpretation and the deterministic many-worlds interpretation. The multiverse tree feature (Figure 3) allows users to investigate the experiment concerning these two interpretations. In the Copenhagen interpretation, all branches are probabilistic outcomes. Within the many-world interpretation, each branch is a coexisting part of the quantum state.

The pre-prepared example setups recreate historical experiments from quantum computing, quantum information, or quantum cryptography and introduce users to some of the interface’s features. Examples include Elitzur-Vaidman bomb tester, BB84 quantum key distribution protocol, Mach-Zehnder and Michelson-Morley interferometers, Deutsch-Jozsa quantum computing algorithm, and quantum teleportation (Figure 1).

In January 2022 we released Quantum Game [17] - a series of puzzles using the Virtual Lab interface and simulation. Levels built as riddles introduce basic concepts of quantum physics, such as the polarization of light, interference, and entanglement. It serves as a casual yet engaging game, a learning material, and an introduction to the Virtual Lab interface.

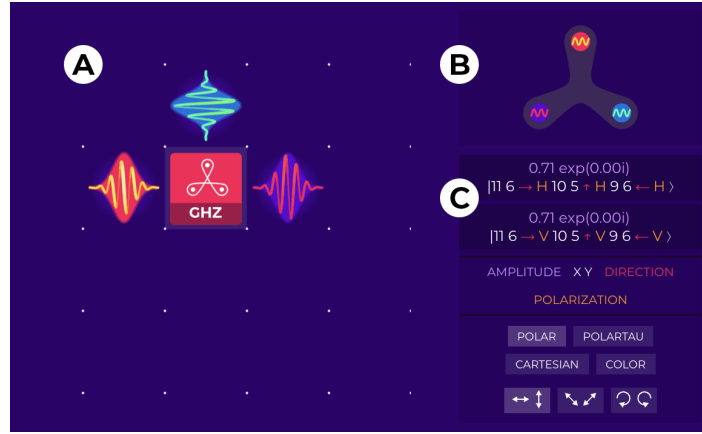


Figure 2: Three ways to investigate entangled particles

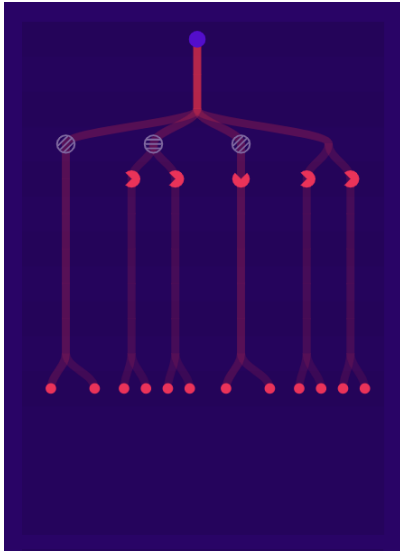


Figure 3: Multiverse Tree feature

4 IMPLEMENTATION

There are many potential implementations of quantum computing, including optically-trapped neutral atoms and ions, superconductors, and photons.

We decided to develop a simulation engine that can be used to simulate the leading implementations of quantum computing. Unlike prevalent qubit-only abstraction, we focus on concrete physical implementations. Since we are now in the era of early-stage quantum devices (Noisy Intermediate Scale Quantum [18]), concrete physical implementations are crucial for understanding capabilities. Moreover, they give a visceral sense of the quantum world.

Classical simulations of quantum systems get exponentially more difficult with the growing number of particles. In our case, a single photon requires 1000 complex numbers (all combinations of its grid position, polarization, and direction). A three-photon simulation requires a billion dimensions. Consequently, Quantum Flytrap needed

to develop a custom high-performance sparse array simulation in Rust, a low-level programming language, as none of the off-the-shelf solutions were suitable. The web interface uses a number of JavaScript technologies, including its dialect TypeScript and the web framework Vue.js.

5 USAGE

University professors and educators use the Virtual Lab as a teaching tool. It has been used during Quantum Information courses at the University of Oxford and Stanford University and for the Introduction to Quantum Computing course at Qubit by Qubit [19]. We heard from one of the professors, whose experimental lab was closed during the COVID pandemic, that “even when it becomes possible to teach my class in person again, your website could help me scale to a larger class size than is possible when I require equipment for hands-on demonstrations.”

The Virtual Lab has been used during several hackathons and other informal educational events and by individual users. As an easily embeddable tool, it has been used by external educational portals for quests and tutorials introducing quantum mechanics and quantum computing [20]. We have approximately 40 unique users per day, growing to 700 during events. There are over 350 user-created experiments, which can be shared. We have created a Discord channel to gather their valuable feedback, cultivate a discussion and develop a new quantum community.

Moreover, the Virtual Lab’s custom-made simulation engine allows us to create user interfaces to interact with actual quantum computers. The simulation will be used for faster prototyping on classical computers. Currently, we work with Pasqal, one of the leading quantum computing manufacturers, developing a user interface for their technology based on neutral Rydberg atoms.

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